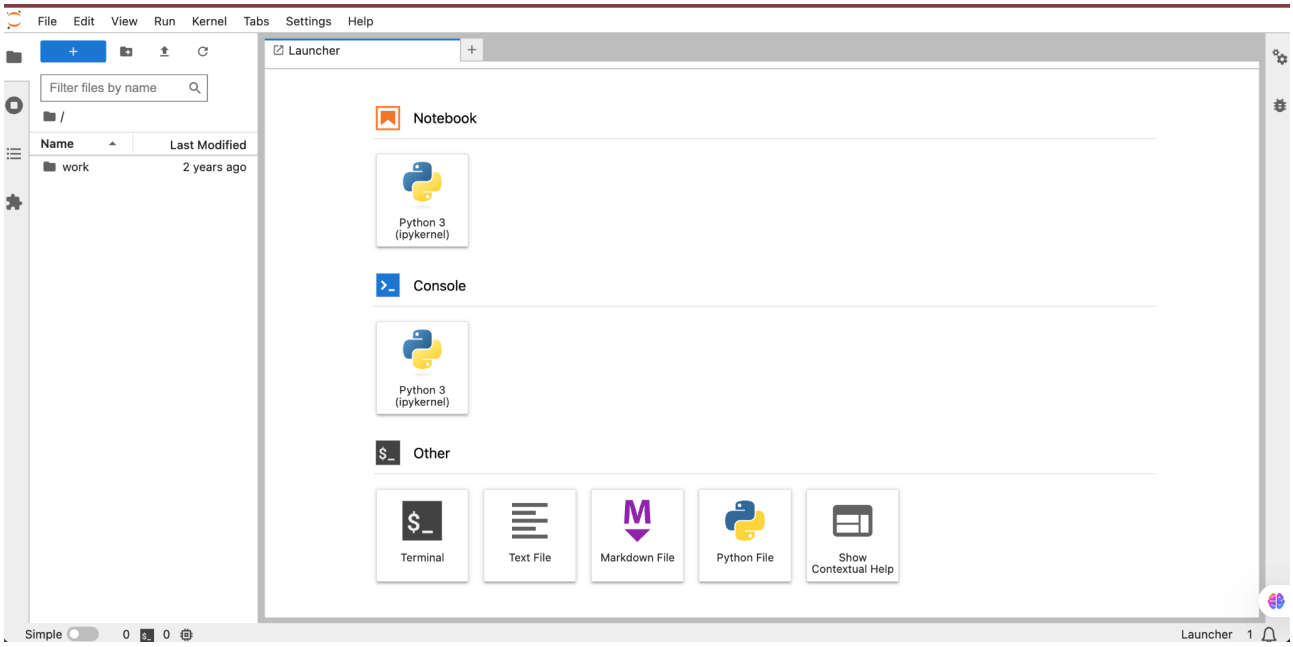
“This interface is JupyterLab running inside Docker. It provides an isolated environment where Apache Spark processes sensor data and machine learning models like LSTM are executed. All data processing, analysis, and visualization are performed here.”

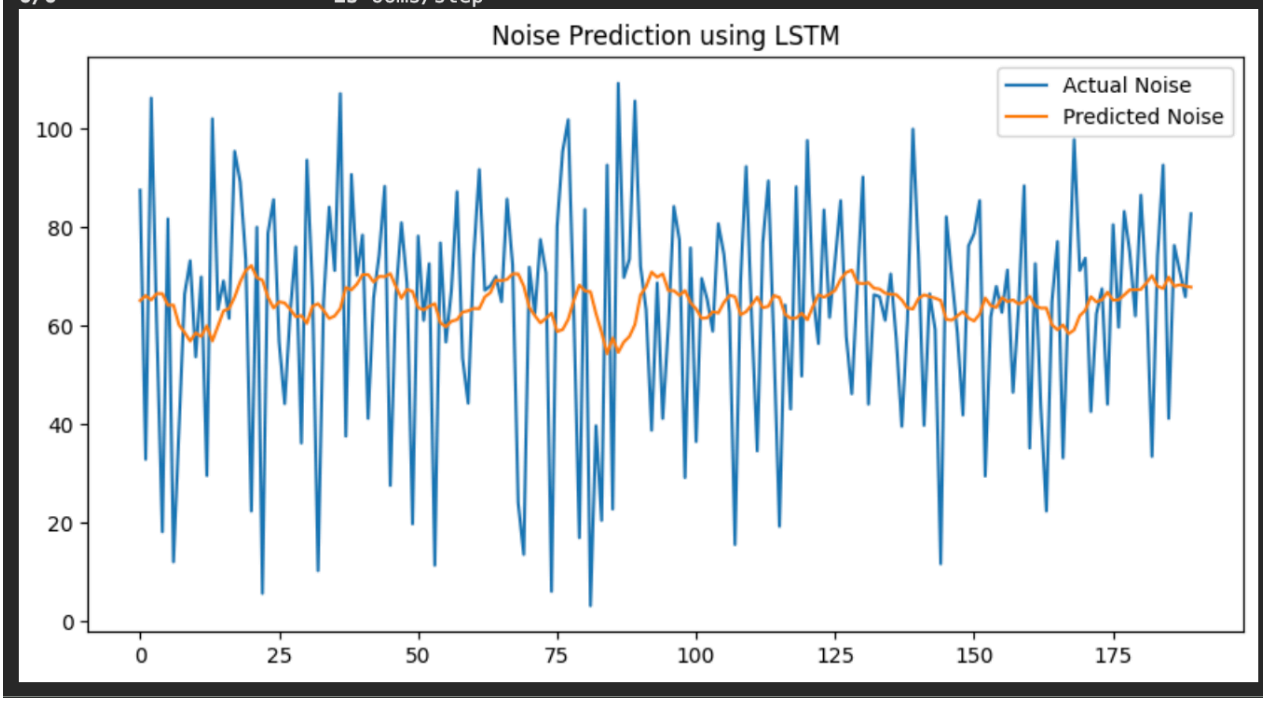



```
pred = model.predict(X)

pred = scaler.inverse_transform(pred)
actual = scaler.inverse_transform(y)

plt.figure(figsize=(10,5))
plt.plot(actual, label="Actual Noise")
plt.plot(pred, label="Predicted Noise")
plt.title("Noise Prediction using LSTM")
plt.legend()
plt.show()
```

```
6/6 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step
```



127.0.0.1:8888/lab/tree/EchoMap_LSTM.ipynb

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EchoMap_...	2 minutes ago
urban-nois...	10 minutes ago

EchoMap_LSTM.ipynb

Code | Open in... | Python 3 (ipykernel)

[8]:
```
print(df.columns)
```

```
Index(['city', 'state', 'country', 'latitude', 'longitude', 'sensor_id',
       'timestamp', 'noise_type', 'reading_id', 'noise_level_db',
       'regulatory_limit_db', 'sensor_location_name',
       'exceeds_regulatory_limit'],
      dtype='object')
```

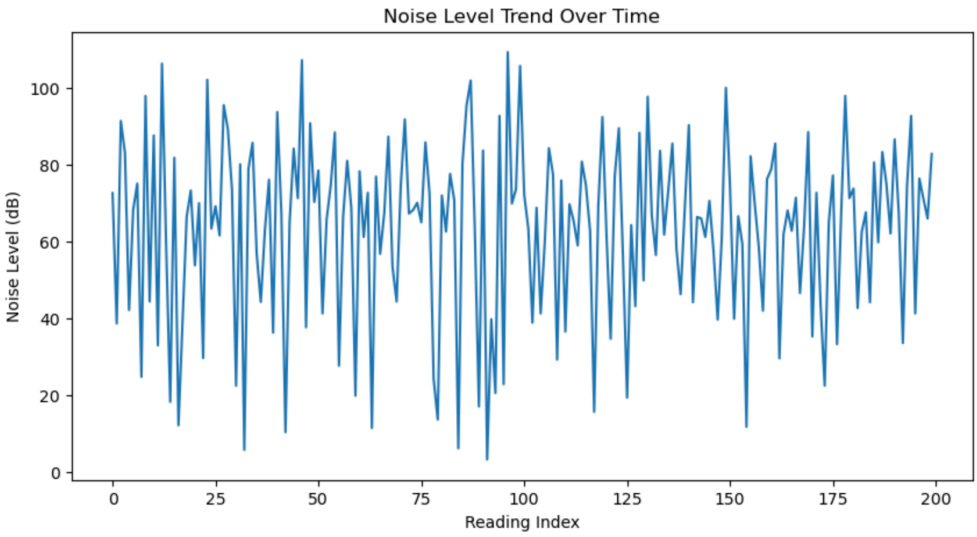
[9]:
```
import pandas as pd

# Load dataset (file already uploaded)
df = pd.read_json('urban-noise-pollution-sensor-data.json')

print("Shape:", df.shape)
df.head()
```

```
Shape: (200, 13)
```

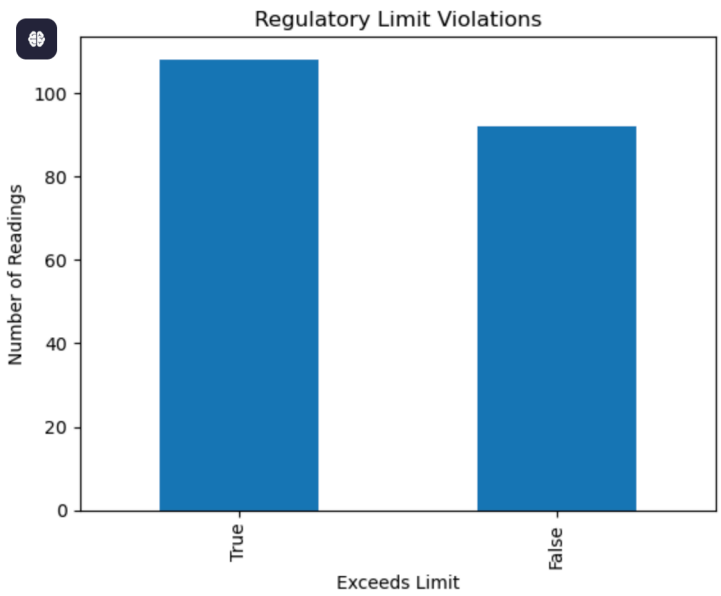
[9]:

	city	state	country	latitude	longitude	sensor_id	timestamp	noise_type	reading_id	noise_level_db	regulatory_limit_db	sen
0	Los Angeles	CA	USA	34.0522	-118.2437	SEN-0001	2024-06-01 00:05:12+00:00	traffic	RD-20240601001	72.5	65	
1	New York	NY	USA	40.7975	-73.9580	SEN-0002	2024-06-01 01:18:34+00:00	nature	RD-20240601002	38.6	45	Cer
2	Chicago	IL	USA	41.8781	-87.6298	SEN-0003	2024-06-01 02:27:10+00:00	crowd	RD-20240601003	91.3	75	
3	Seattle	WA	USA	47.6062	-122.3321	SEN-0004	2024-06-01 03:45:19+00:00	construction	RD-20240601004	83.0	80	Ha
4	Huntington Beach	CA	USA	33.7205	-118.0806	SEN-0005	2024-06-01 04:02:41+00:00	nature	RD-20240601005	42.1	50	

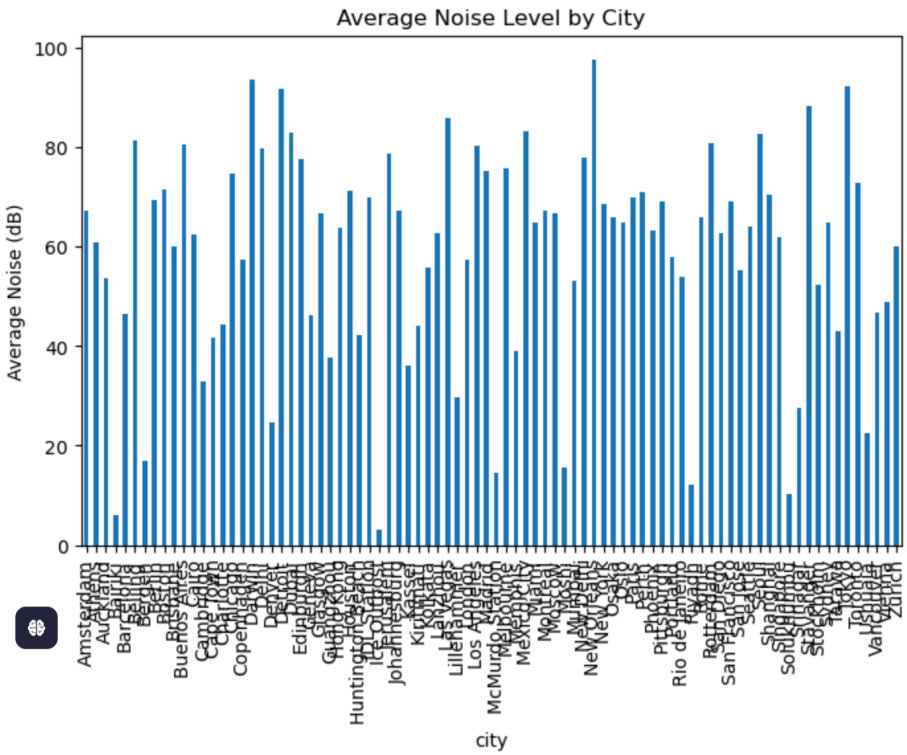
[10]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(df['noise_level_db'])
plt.title("Noise Level Trend Over Time")
plt.xlabel("Reading Index")
plt.ylabel("Noise Level (dB)")
plt.show()
```



[12]:
```
df['exceeds_regulatory_limit'].value_counts().plot(kind='bar')
plt.title("Regulatory Limit Violations")
plt.xlabel("Exceeds Limit")
plt.ylabel("Number of Readings")
plt.show()
```



[13]:
```
df.groupby('city')['noise_level_db'].mean().plot(kind='bar', figsize=(8,5))
plt.title("Average Noise Level by City")
plt.ylabel("Average Noise (dB)")
plt.show()
```



A regression model was used to predict noise levels based on location, validating the analytical capability of the system.

[14]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X = df[['latitude', 'longitude']]
y = df['noise_level_db']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)

print("Model Accuracy:", model.score(X_test, y_test))
```

```
Model Accuracy: 0.09778908156275246
```

[ ]: